

ARTIFICIAL

NEURAL

NETWORKS (ANN)

AND DEEP

Machine Learning Technical Presentation

LEARNING

Name: Rishwanth. L

Register No.: 192424100



Guided By:

Dr. R. Kavitha

MACHINE LEARNING & DEEP LEARNING

- Machine Learning learns patterns from data.
- Deep Learning is a subset of ML using Artificial Neural Networks.
- Deep Learning automatically extracts features.

Machine Learning (ML)

- ML enables computers to learn patterns from data without explicit programming.
- Requires manual feature selection.
- Suitable for structured data.

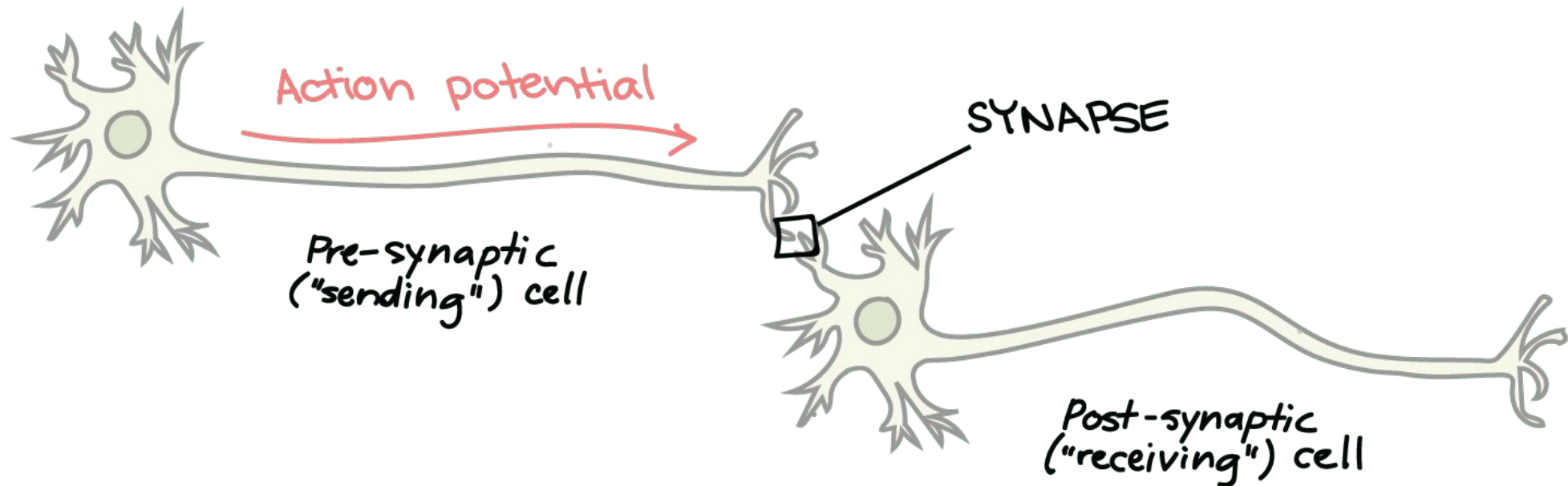
Deep Learning (DL)

- A subset of Machine Learning.
- Uses Artificial Neural Networks with multiple hidden layers.
- Automatically learns important features from data.

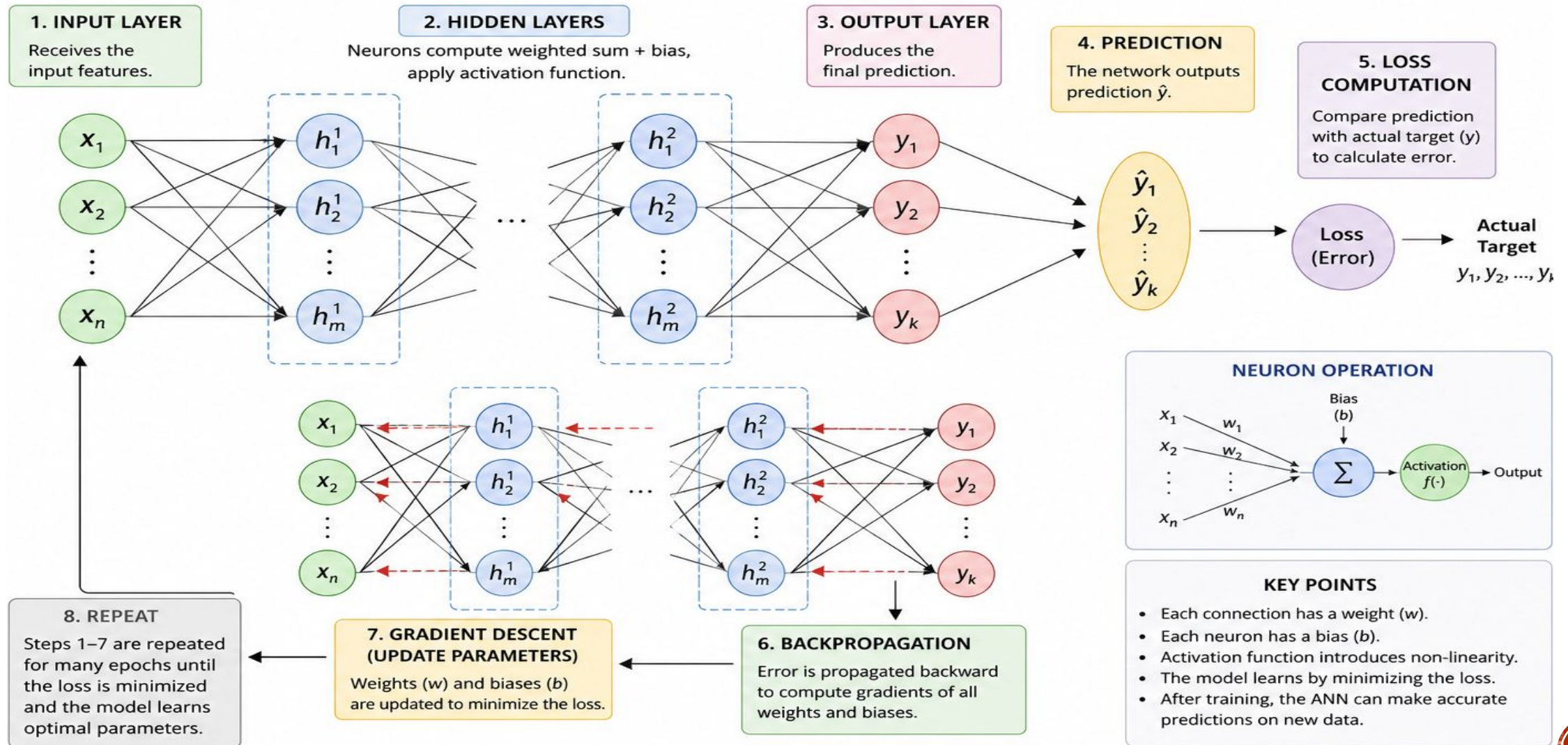


ARTIFICIAL NEURAL NETWORK (ANN)

- An Artificial Neural Network (ANN) is a computational model inspired by the human brain. It consists of interconnected neurons that process information and learn patterns from data.
- It consists of Input Layer, Hidden Layer(s) and Output Layer.
- Neurons are connected through weights and bias.



HOW ANN LEARNS



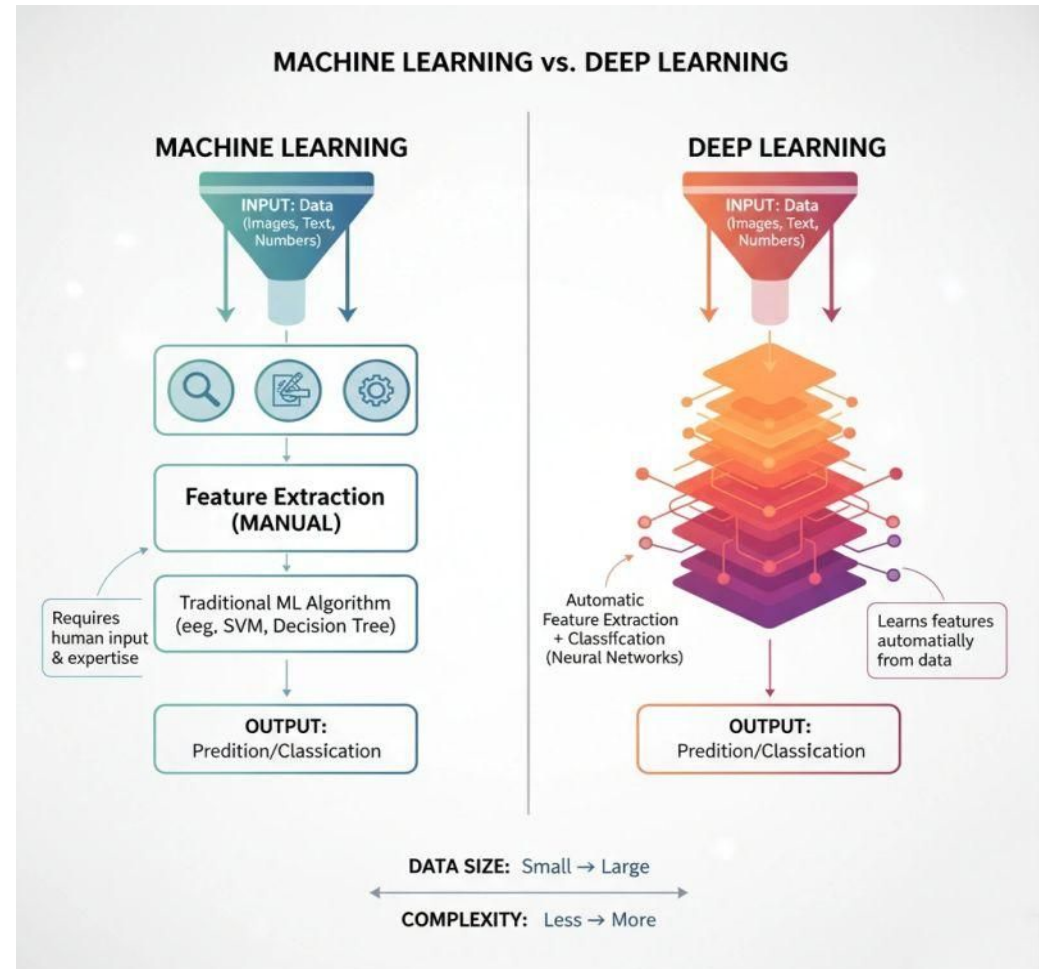
ACTIVATION FUNCTIONS

Activation Function	How it Works	Output Range	Advantages	Common Applications
Sigmoid	Converts any input into a value between 0 and 1, making it suitable for representing probabilities.	0 to 1	Produces probability-like outputs, easy to interpret.	Binary classification (Spam/Not Spam, Disease/No Disease).
ReLU (Rectified Linear Unit)	Outputs the input directly if it is positive; otherwise outputs 0.	0 to infinity	Fast computation, reduces the vanishing gradient problem, and speeds up training.	Most widely used activation function in hidden layers of deep neural networks.
Tanh (Hyperbolic Tangent)	Similar to Sigmoid but maps values between -1 and 1, allowing both positive and negative outputs.	-1 to 1	Zero-centered outputs help networks converge faster than Sigmoid.	Hidden layers in recurrent neural networks (RNNs) and sequence models.
Softmax	Converts multiple output values into probabilities that sum to 1. The highest probability determines the predicted class.	0 to 1 (sum = 1)	Provides probability distribution across multiple classes.	Multi-class classification (Digit Recognition, Image Classification, Language Detection).



DEEP LEARNING VS TRADITIONAL ML

- ML requires manual feature engineering.
- DL automatically learns features.
- DL performs better on images, audio and text.
- DL benefits from large datasets and GPUs.



APPLICATIONS

- Healthcare and disease diagnosis
- Face recognition
- ChatGPT and virtual assistants
- Self-driving cars
- Fraud detection and recommendation systems



CONCLUSION

- ANN is the foundation of Deep Learning.
- Backpropagation and Gradient Descent enable learning.
- Deep Learning powers modern AI applications.





THANK YOU